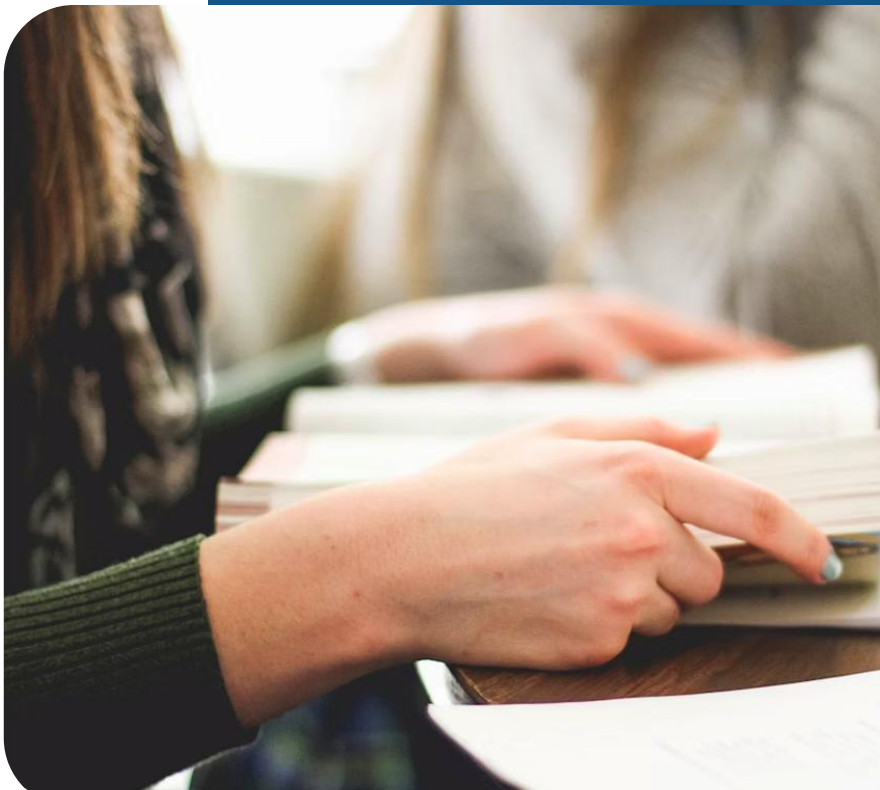
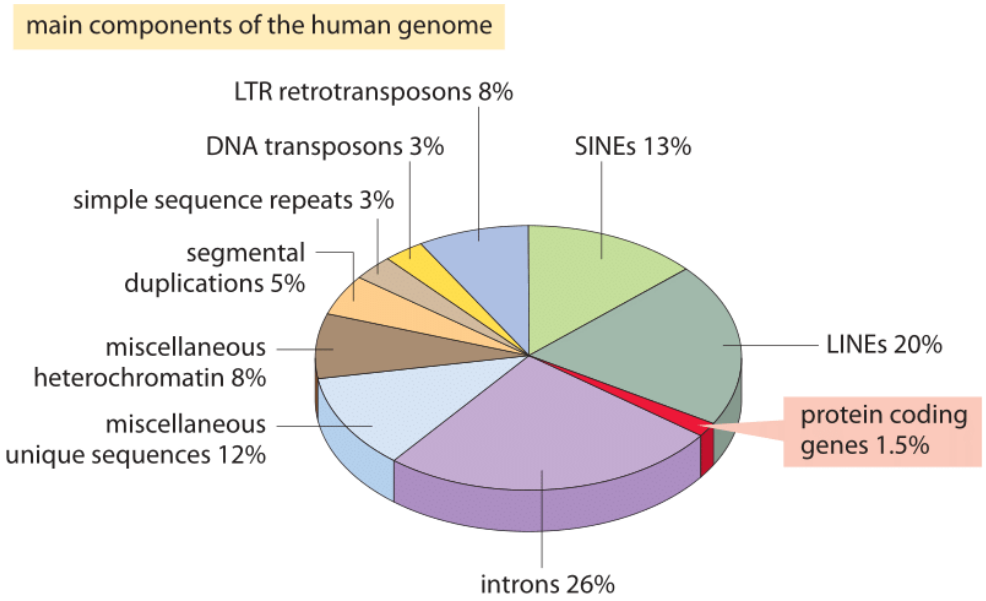
DNA
Mutation





Only a small fraction of the human genome directly codes for proteins, showing that DNA is far more than just a collection of genes.

Large portions of the genome, such as introns and regulatory sequences, play critical roles in controlling gene activity, RNA processing, and expression patterns.

Other components, including repetitive elements and structural regions, contribute to genome stability, organization, and long-term evolution.

Read other component in DNA beside protein coding gene

A MUTATION:

A CHANGE IN THE DNA BASE SEQUENCE

HAPPEN SPONTANEOUSLY IN OUR CELLS

(WHEN DNA IS DUPLICATED)

INCREASE THE RISK:

1. CARCINOGENS



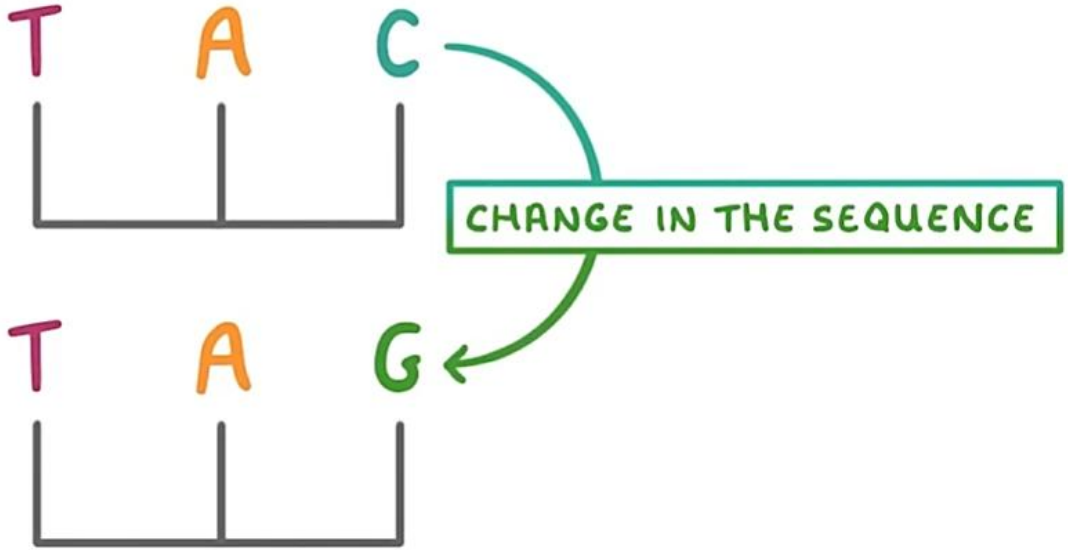
HARMFUL CHEMICALS

2. CERTAIN TYPES OF RADIATION









HOW DNA WORKS:

MOST OF THE TIME:

A MUTATION → DOESN'T HAVE ANY SIGNIFICANT EFFECT

STILL WORKS IN THE SAME WAY



G A T A G T A C C T A G C T A G C A G A C

A MUTATION → MOST OCCUR



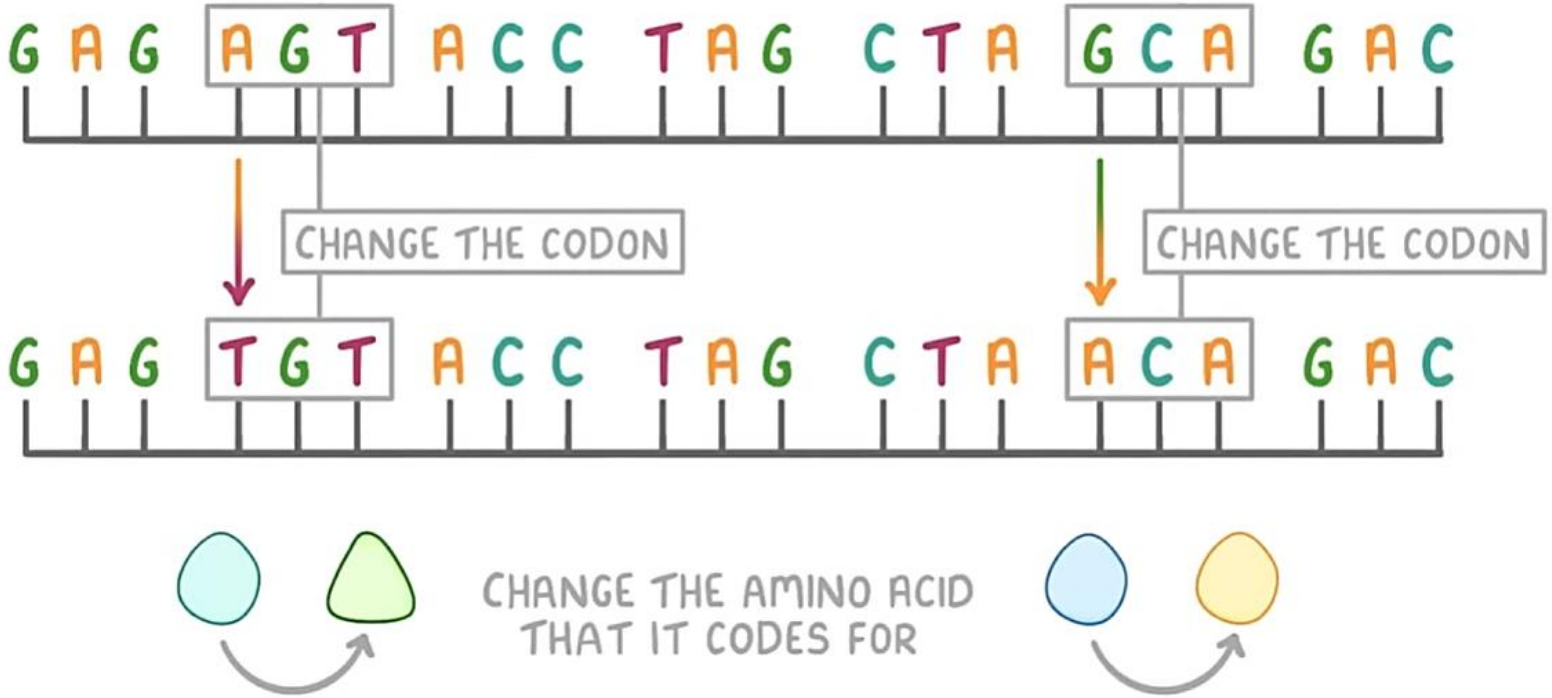
NON-CODING DNA

(DOESN'T CODE FOR A PROTEIN)

IMPORTANT ROLE IN THE EXPRESSION OF GENES

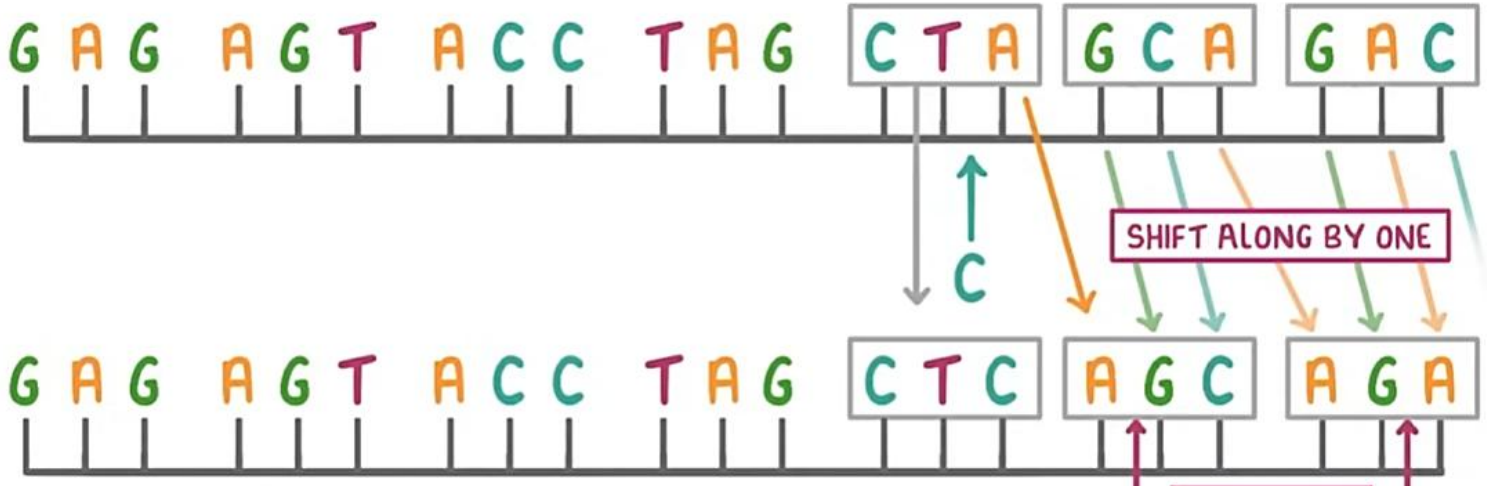
SUBSTITUTIONS

ONE OF THE BASES IS CHANGED FOR ANOTHER RANDOM BASE



INSERTION

AN EXTRA BASE IS INSERTED INTO THE SEQUENCE



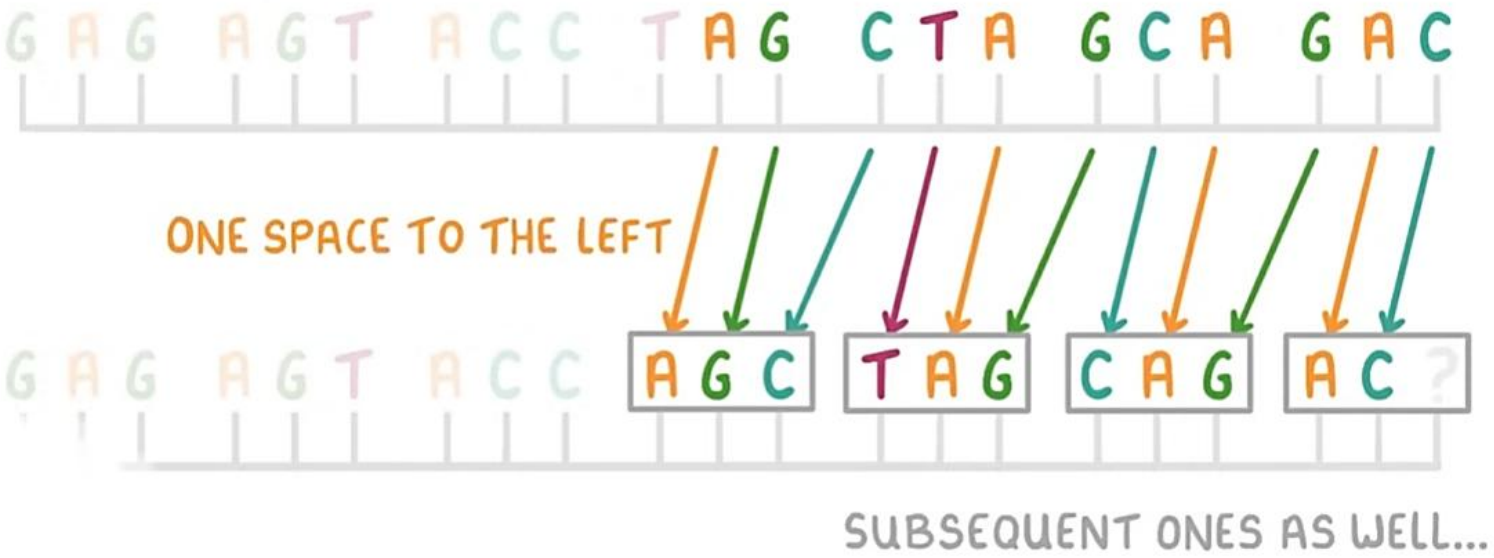
AMINO ACID CHAIN WOULD BE COMPLETELY DIFFERENT

WORSE THAN

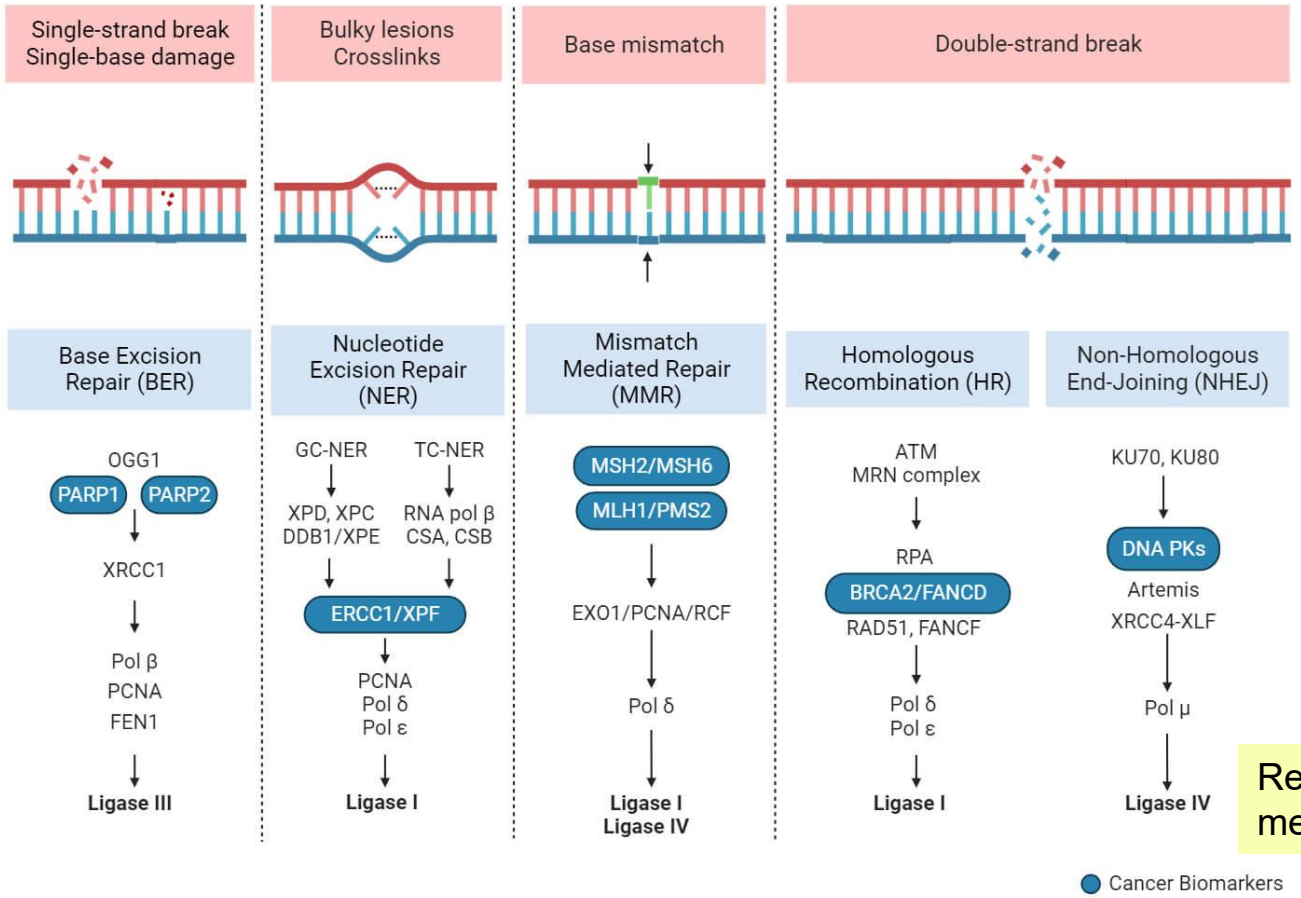
SUBSTITUTIONS

DELETION

ONE OF THE BASES IS DELETED FROM THE SEQUENCE

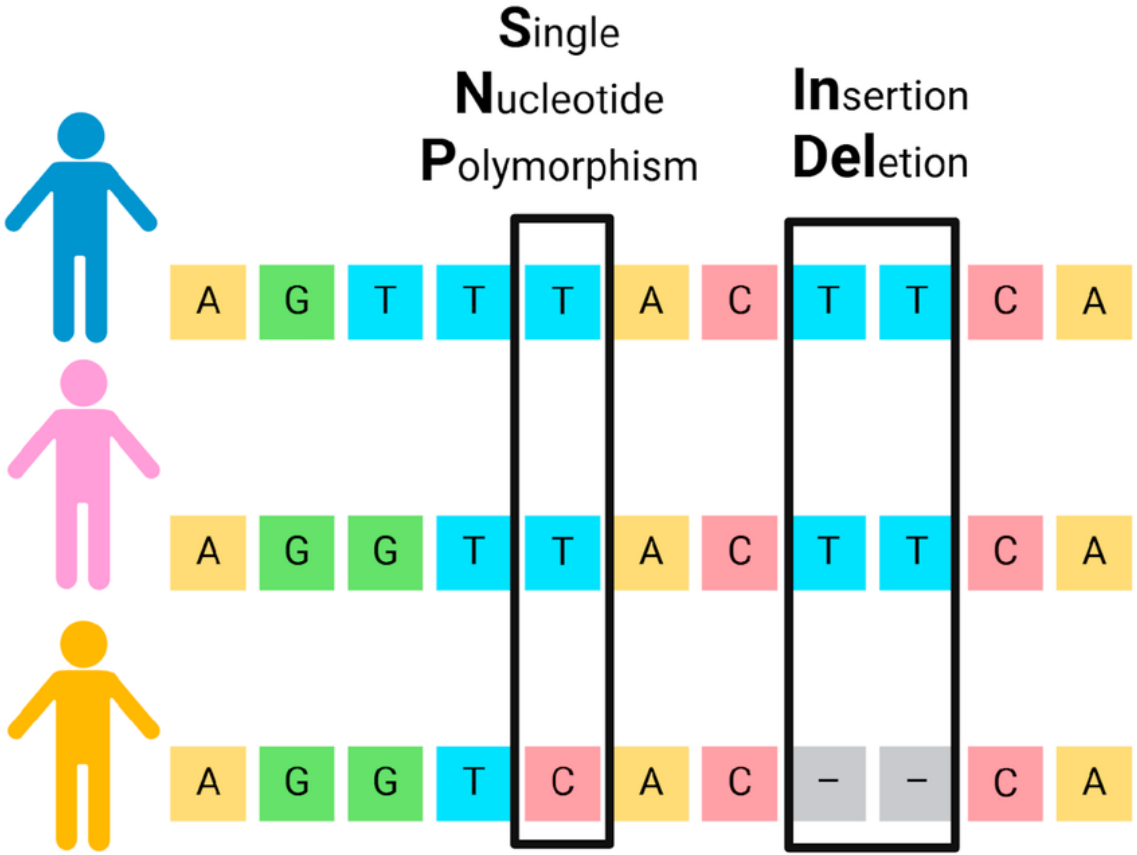


DNA Repair Mechanisms



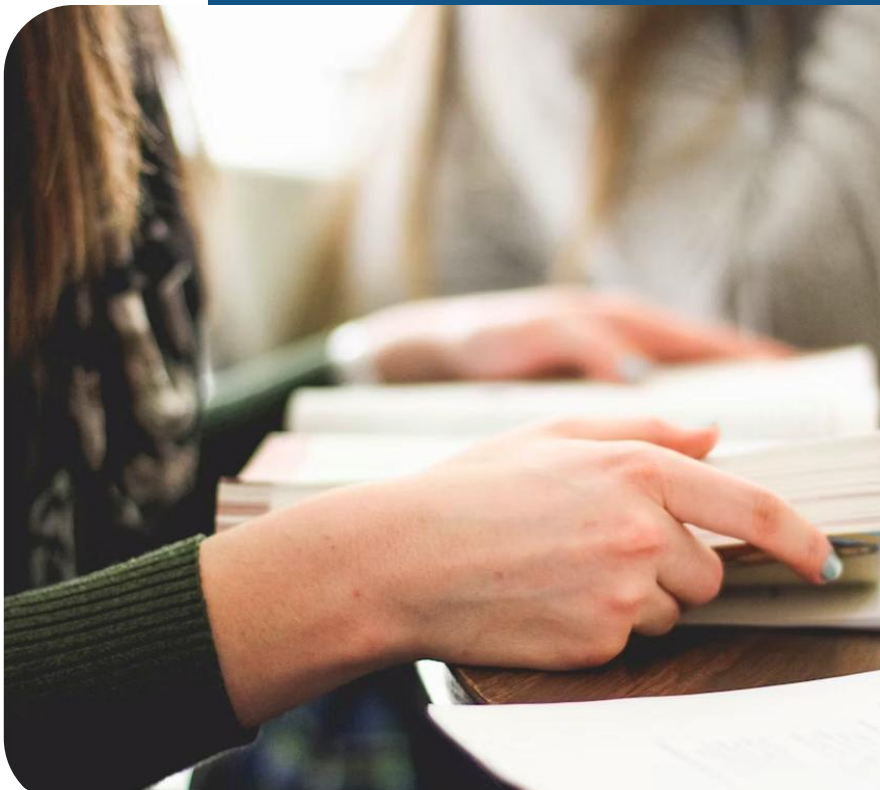
Read DNA repair mechanism

Cancer Biomarkers



Polymorphism, derived from Greek meaning "many forms," refers to the ability of an entity—such as an object, variable, or method—to take on multiple forms or behaviors

DNA
Sequencing



Computational Biology -Tech

Now that we understand how DNA changes through mutation and evolution, an important question emerges:

How do we actually read, copy, and analyze DNA in the real world?

In computational biology, our models and algorithms do not operate on abstract molecules — they rely on data generated by powerful laboratory technologies. Techniques such as **DNA sequencing** and **PCR** are what transform biological molecules into digital information that we can store, process, and study.

DNA Sequencing

DNA sequencing refers to methods for determining the order of the nucleotide bases adenine, guanine, cytosine, and thymine in a molecule of DNA.

The first DNA sequence was obtained by academic researchers, using laboratories methods based on 2- dimensional chromatography in the early 1970s.

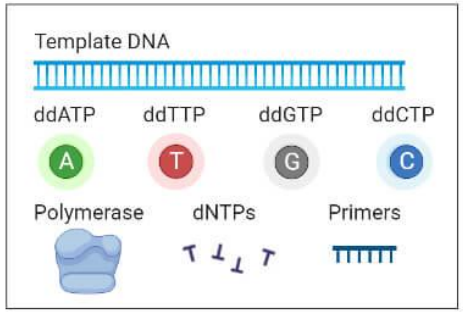
Two main methods are widely known to be used to sequence DNA:

- The Chemical Method (also called the Maxam–Gilbert method after its inventors).
- The Chain Termination Method (also known as the Sanger dideoxy method after its inventor).

Sanger Sequencing

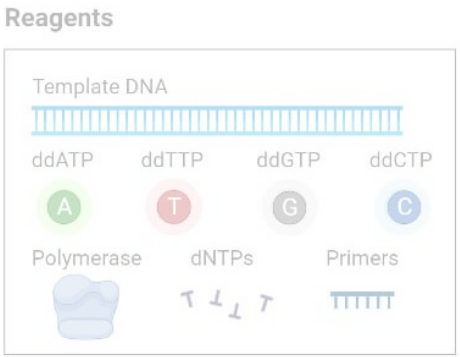
Microbe Notes
The Biology Notes
The Chemistry Notes
Created with bio RENDER
Designed By Sagar Aryal

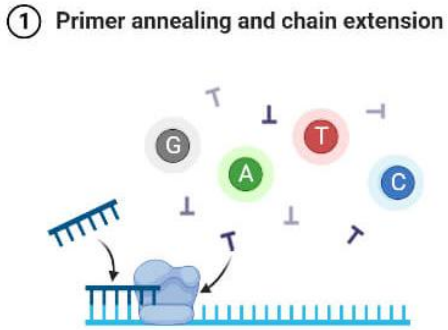
Reagents



- The key principle of the Sanger method was the use of **dideoxynucleotide triphosphates (ddNTPs)** as **DNA chain terminators**.
- The chain termination method requires a single-stranded DNA template, a DNA primer, a DNA polymerase, free nucleotide bases and radioactively or fluorescently labelled nucleotides that terminate DNA strand elongation.

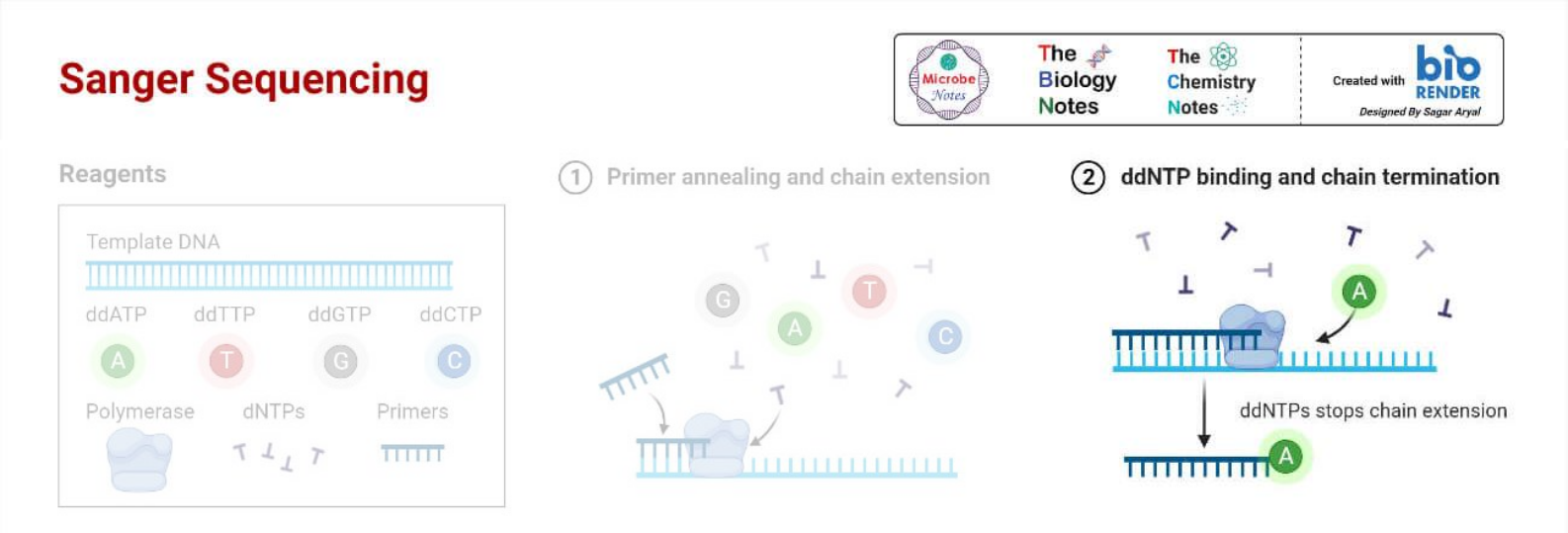
Sanger Sequencing





- The DNA sample is divided into four separate sequencing reactions*, containing all four of the standard deoxynucleotides (dATP, dGTP, dCTP, dTTP) and the DNA polymerase.

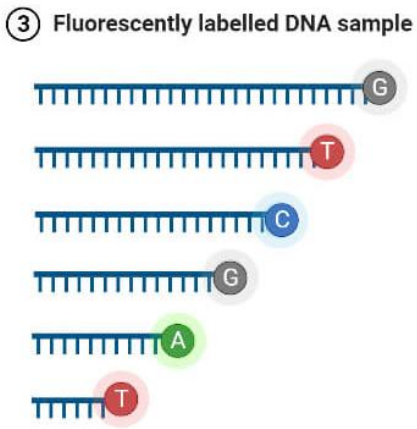
*For Manual enzymatic DNA sequencing

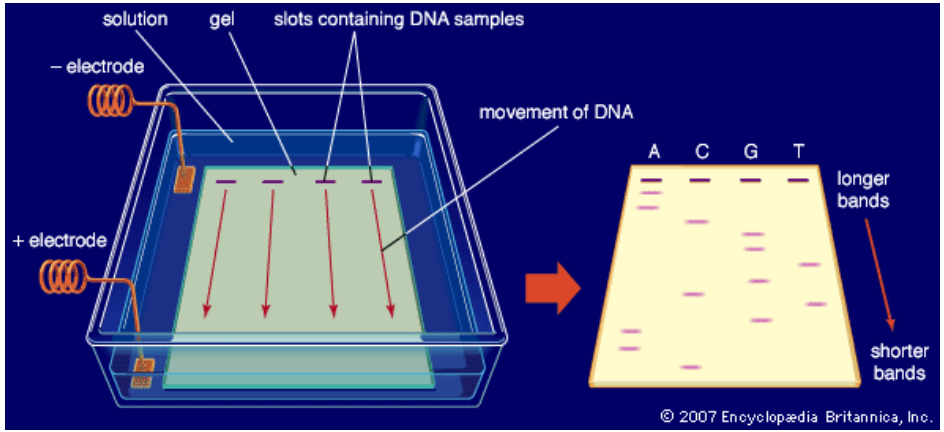


- To each reaction is added only one of the four dideoxynucleotide (ddATP, ddGTP, ddCTP, ddTTP) which are the chain terminating nucleotides, lacking a 3'-OH group required for the formation of a phosphodiester bond between two nucleotides, **thus terminating DNA strand extension** and resulting in DNA fragments of varying length.

Sanger Sequencing

- The newly synthesized and labelled DNA fragments are heat denatured, and separated by size by gel electrophoresis on a denaturing polyacrylamide-urea gel with each of the four reactions run in one of the four individual lanes (lanes A, T, G, C).



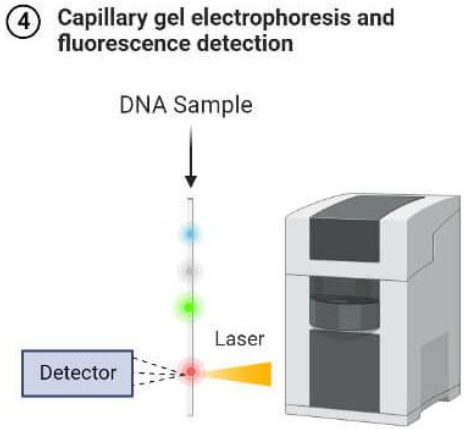


Sanger Sequencing

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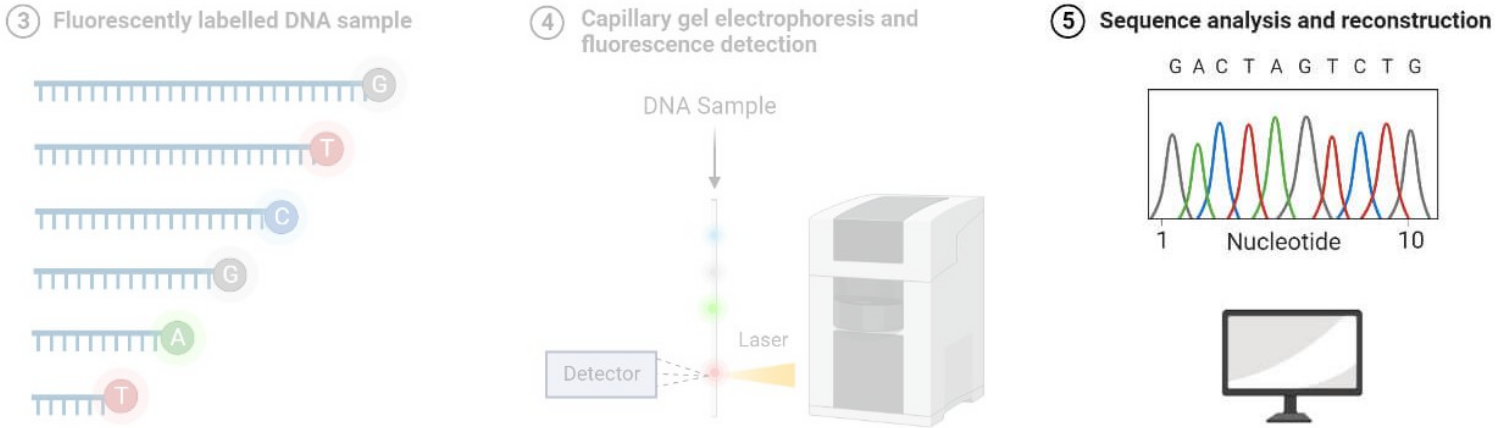
- The fragment mix is injected into a thin capillary filled with a polymer (acts like a gel). An electric field pulls DNA through the capillary. Smaller fragments move faster → so they arrive first.





Sanger Sequencing

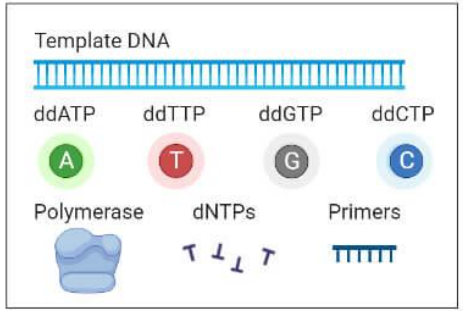
- Converts the signal into a chromatogram (colored peaks). Calls bases (A/T/G/C) in order of fragment arrival: shortest → longest. Assigns quality scores (confidence per base)



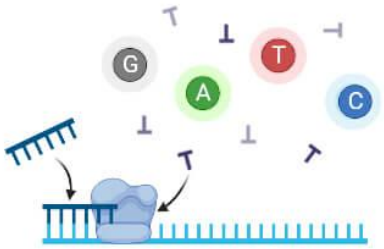
Sanger Sequencing

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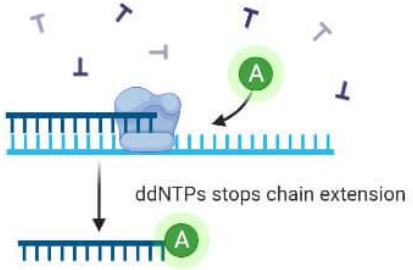
Reagents



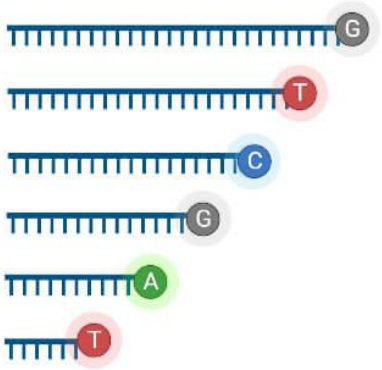
1 Primer annealing and chain extension



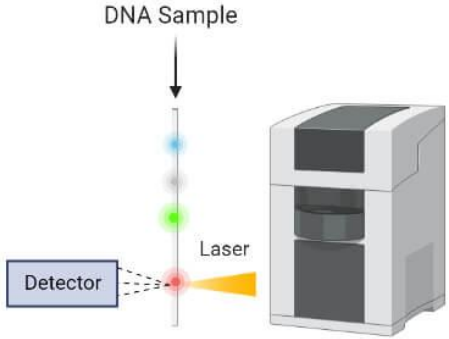
2 ddNTP binding and chain termination



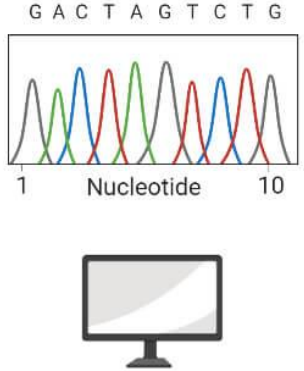
3 Fluorescently labelled DNA sample

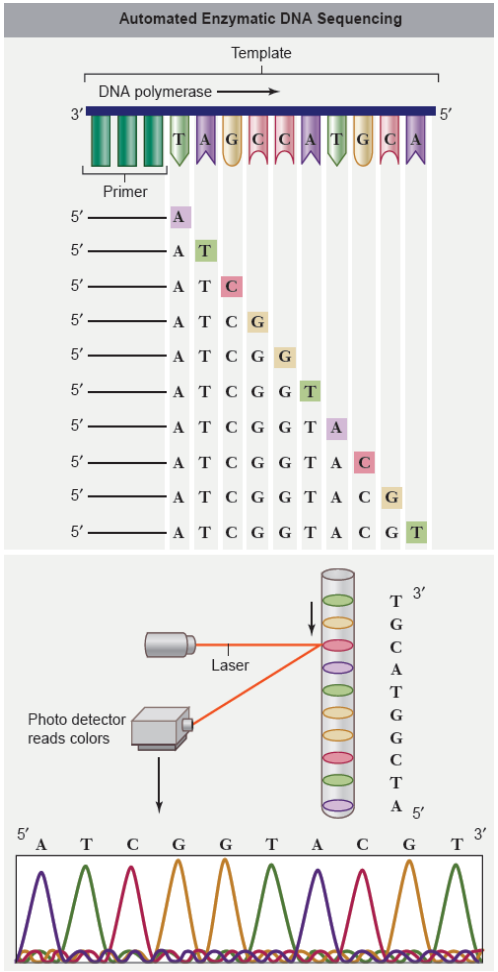


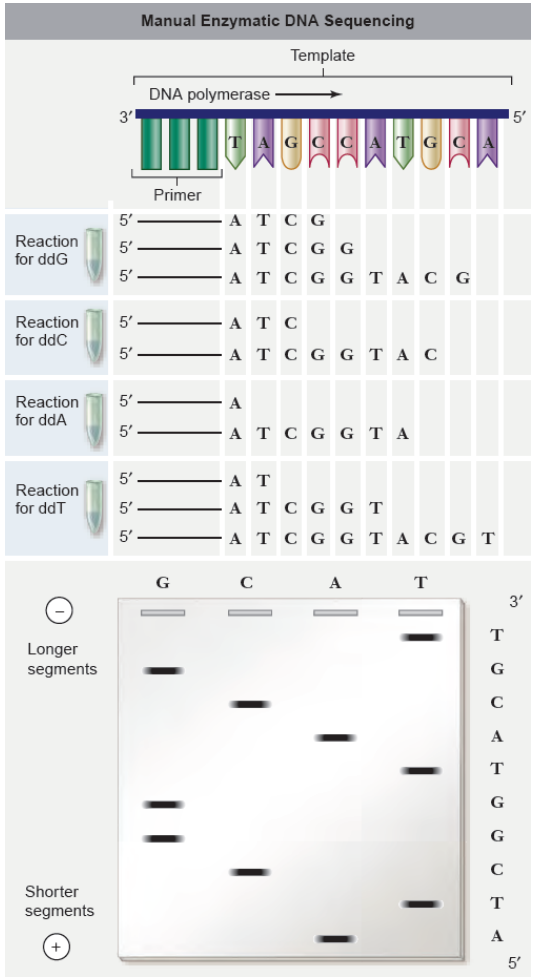
4 Capillary gel electrophoresis and fluorescence detection



5 Sequence analysis and reconstruction







Manual Enzymatic DNA Sequencing:

The DNA bands are visualized by autoradiography or UV light, and the DNA sequence can be directly read off the X-ray film or gel image.

A dark band in a lane indicates a DNA fragment that is result of chain termination after incorporation of a dideoxynucleotide (ddATP, ddGTP, ddCTP, or ddTTP).

The relative position of the different bands among the four lanes are then used to read (from bottom to top) the DNA sequence.

Sanger Method - Typical performance

Read length: ~**600–900 bp** (sometimes longer with ideal templates)

Accuracy: very high in the middle of the read; usually worse at the start and especially toward the end (peaks overlap as fragments get long)

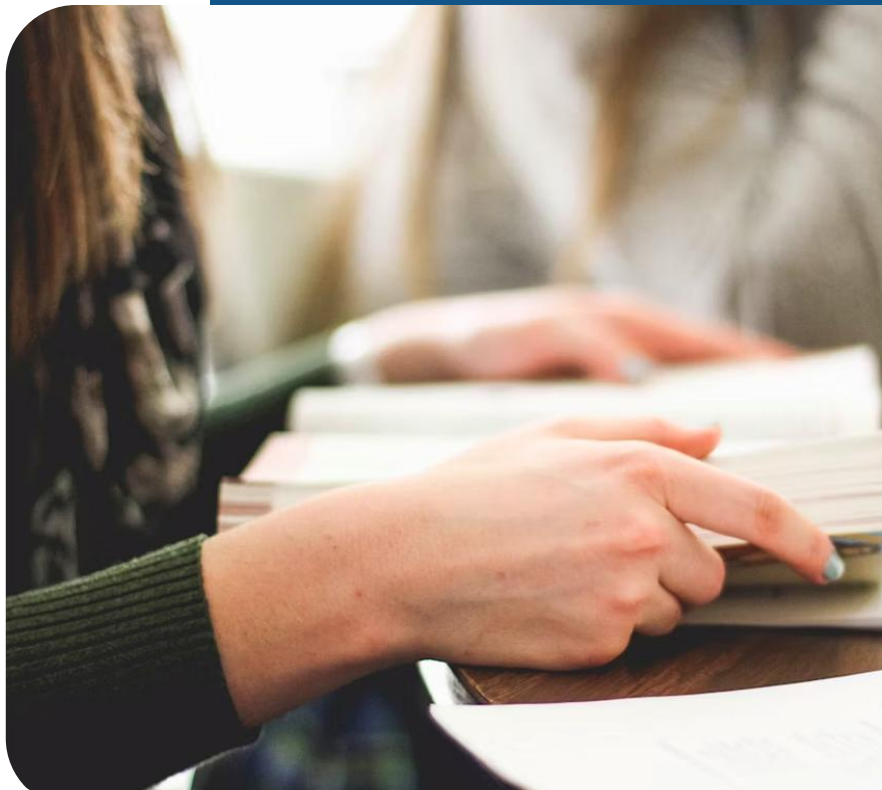
Sanger is best for: single genes, plasmids, confirming a mutation, validating NGS results.

Limitations: not efficient for whole genomes; struggles with mixed templates (multiple sequences → messy overlapping peaks).

Other Methods

Feature	Sanger	NGS (Illumina)	Long-Read (ONT / PacBio)
Accuracy per read	Very high	Very high	Improving / high
Read length	~500–1000 bp	Short (50–300 bp)	Very long (kb–Mb)
Throughput	Very low	Massive	Moderate–high
Cost per base	Expensive	Very cheap	Decreasing
Best use cases	Small targets, validation	Whole genomes, transcriptomics	Assembly, structural variation

Polymerase Chain Reaction (PCR)



Early Sanger sequencing

Early Sanger sequencing did **not rely on PCR**.

Instead, it required:

- Relatively **large amounts of purified DNA**
- Often **cloned DNA fragments** inserted into bacteria (plasmids)
- Primer-based extension with polymerase + ddNTP termination

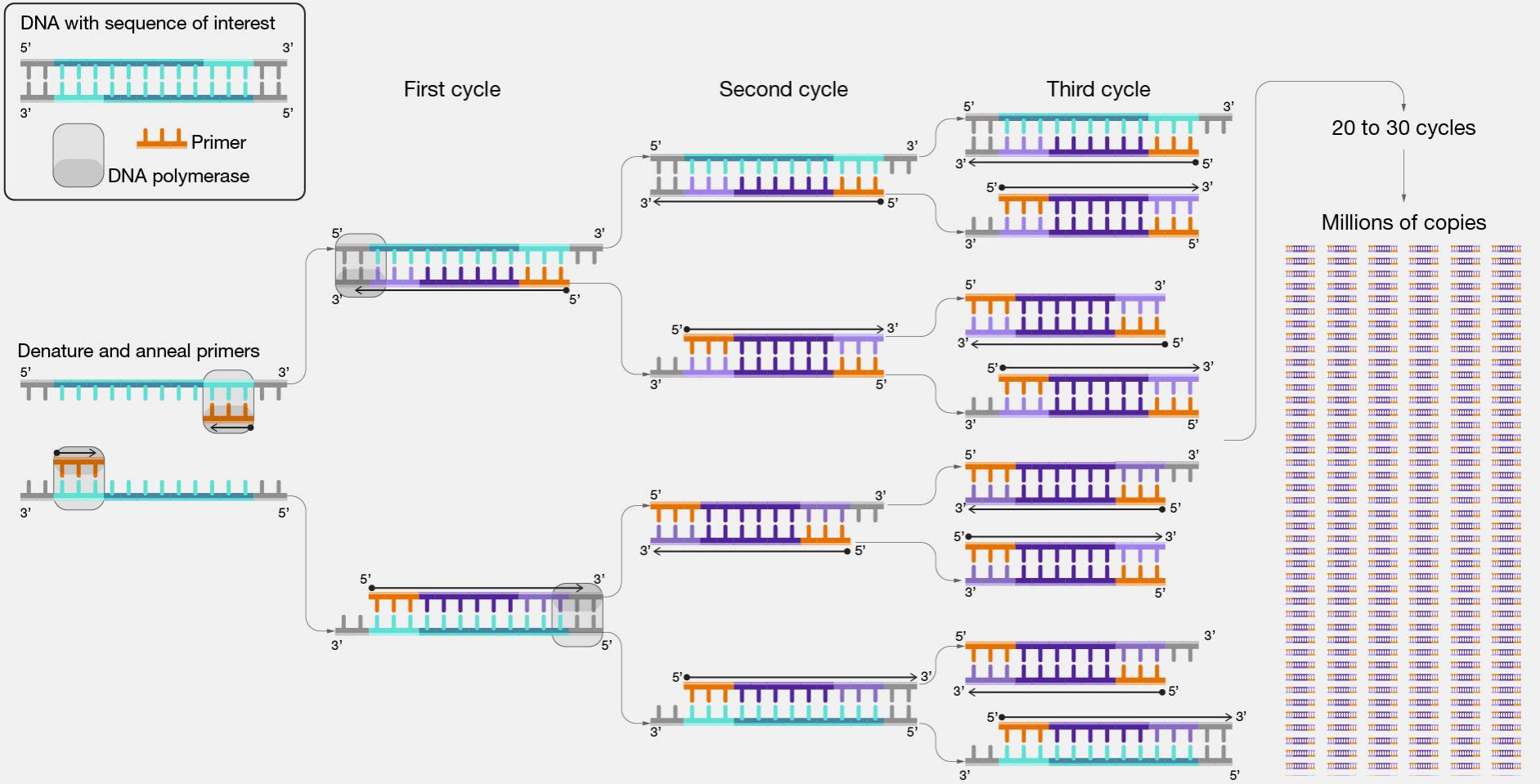
This made sequencing:

- Slower
- More labor-intensive
- Dependent on molecular cloning

Polymerase chain reaction

Polymerase chain reaction (abbreviated PCR) is a laboratory technique for rapidly producing (amplifying) millions to billions of copies of a specific segment of DNA, which can then be studied in greater detail.

PCR involves using short synthetic DNA fragments called primers to select a segment of the genome to be amplified, and then multiple rounds of DNA synthesis to amplify that segment.


DNA with sequence of interest
5'
3'
3'
5'
Primer
DNA polymerase
First cycle
Second cycle
Third cycle
Denature and anneal primers
20 to 30 cycles
Millions of copies

	Senin	Selasa	Rabu	Kamis	Jumat	Sabtu	Minggu
FEB	16	17	18	19	20	21	22
	23	24	25	26	27	28	1
MAR	2	3	4	5	6	7	8
	9	10	11	12	13	14	15
	16	17	18	19	20	21	22
	23	24	25	26	27	28	29
APR	30	31	1	2	3	4	5
	6	7	8	9	10	11	12



Move Session 6 – **14 Mar** to **28 Feb**

THANK YOU!